

**Lab Report No _3_
Memory Read and Write operations**

Microprocessor and Interfacing techniques



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Dated:

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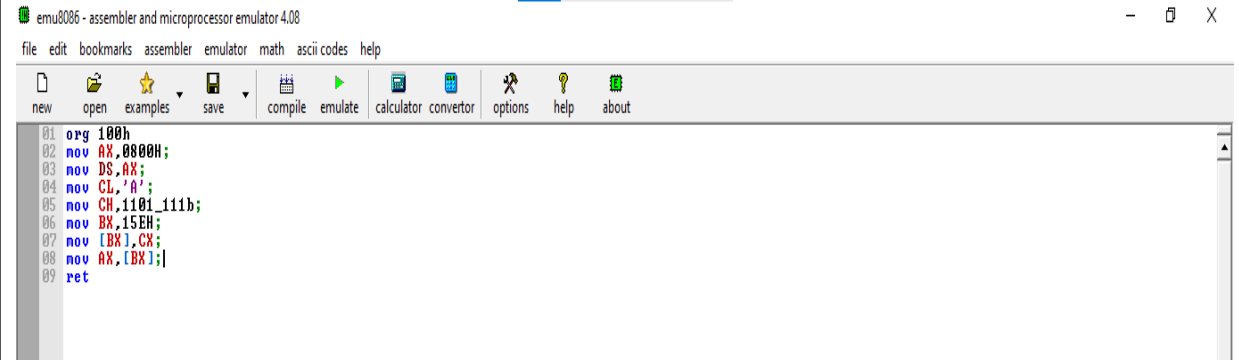
Lab Task No 1:

Solution:

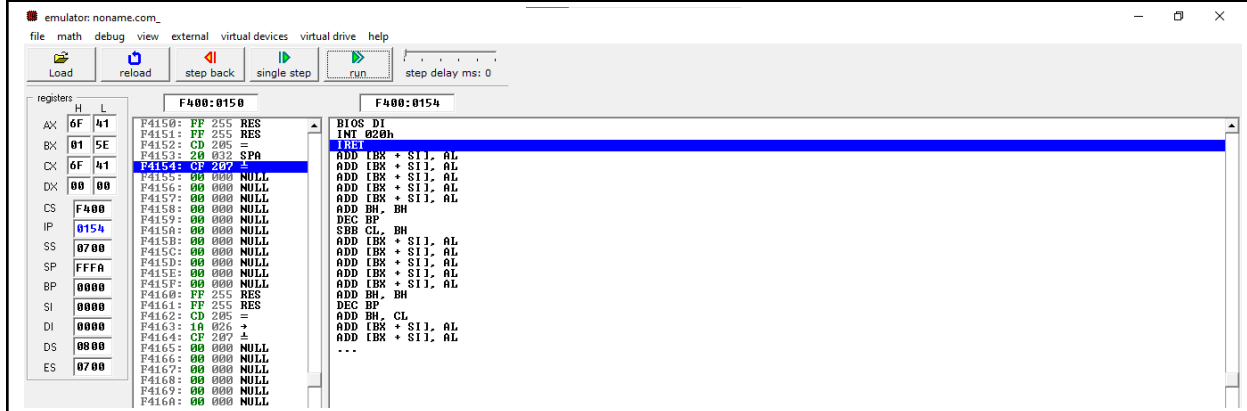
Brief description (3-5 lines)

Run the code and analyze the output.

The code



The results (Screenshot)



Lab Task No 2:

Solution:

Brief description (3-5 lines)

20 bit address =081E0 and DS=0810

The code

```

01 org 100h
02 MOV AX, 0810h
03 MOV DS, AX
04 MOV SI, 0E0h
05 MOV DI, 0810h
06 HLT
07 DW 0x0A55
08 ret

```

The results (Screenshot)

Register	H	L
AX	08	10
BX	00	00
CX	00	0F
DX	00	00
CS	0700	
IP	0100	
SS	0700	
SP	FFFE	
BP	0000	
SI	00E0	
DI	0810	
DS	0810	
ES	0700	

Address	Instruction
07100: B8 10 08	MOV AX, 0810h
07101: 10 01 07	MOV DS, AX
07102: 00 00 07	MOV SI, 0000h
07103: 8E 10 07	MOV DI, 0810h
07104: 00 00 07	HIT
07105: 0E 10 07	PUSH BP
07106: E0 22 07	STOSB
07107: 00 00 07	RET
07108: 00 00 07	NOP
07109: 00 00 07	NOP
07110: 00 00 07	NOP
07111: 00 00 07	NOP
07112: 00 00 07	NOP
07113: 00 00 07	NOP
07114: 00 00 07	NOP
07115: 00 00 07	NOP
07116: 00 00 07	NOP
07117: 00 00 07	NOP
07118: 00 00 07	NOP
07119: 00 00 07	NOP

Lab Task No 3:

Solution:

Brief description (3-5 lines)

Move 01FH into AL register and then add AL register value with 01H. And again add al register value with 0E0H and check the flag status.

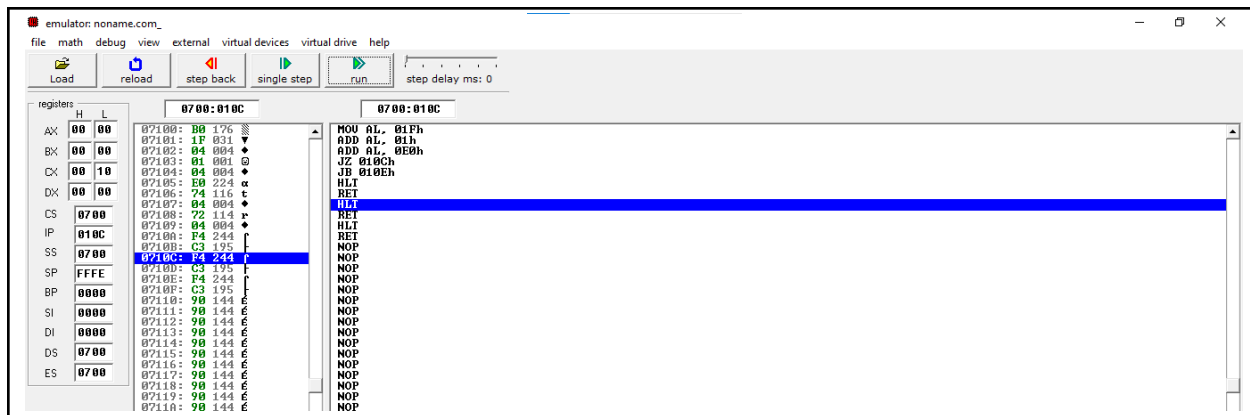
The code

```

01 org 100h
02 MOV AL, 01FH
03 ADD AL, 01H
04 ADD AL, 0E0H
05 JZ ZeroFlagSet
06 JC CarryFlagSet
07 HLT
08 RET
09 ZeroFlagSet:
10 HLT
11 RET
12 CarryFlagSet:
13 HLT
14 ret

```

The results (Screenshot)



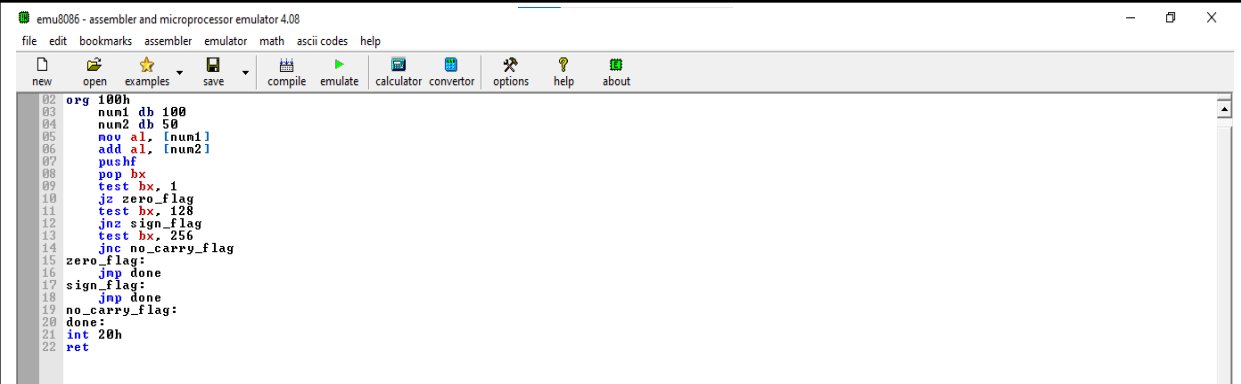
Lab Task No 4:

Solution:

Brief description (3-5 lines)

Add the two numbers 100 or 50 and analyzes the flags.

The code



The results (Screenshot)

